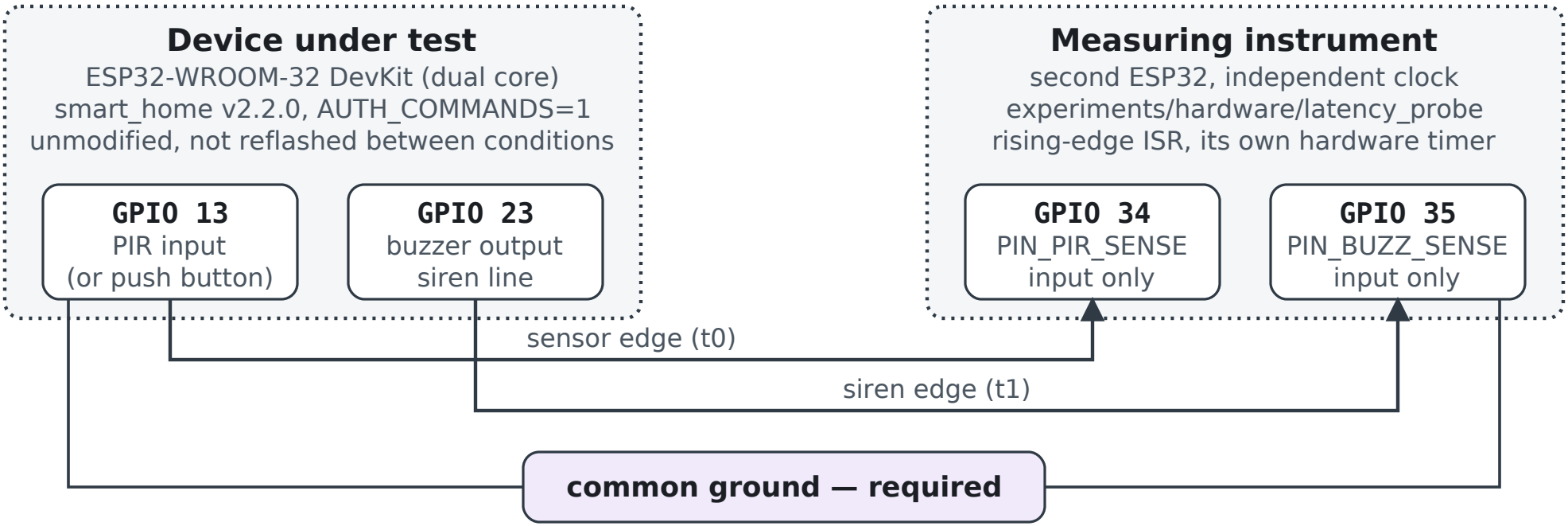
NOT YET BUILT AND NOT YET RUN. No number in the manuscript comes from this rig.

The device under test never times itself: the hypothesis concerns its own clock under load, so a second board with an independent clock timestamps both edges.



Measured quantity: L = t1 − t0, the confirmed sensor edge to the siren line going active

One value per trial, in microseconds, timed entirely on the probe's clock. Written to results_hw.csv by collect.py.

Four conditions, same firmware throughout

A.	idle, broker reachable, no other traffic — the control	none
B.	legitimate MQTT load: frequent STATUS, large retained payloads	none
C.	high-rate publication on the device's own command topic	none
D.	802.11 deauthentication and disassociation frames at the device	shielded enclosure + permission

At least 50 samples per condition, 200 preferred: 200 detects a shift of about 0.29 standard deviations at the pre-registered two-sided Mann–Whitney U test, $\alpha = 0.05$, power = 0.80.